

Day 2: EC2 Instances & Web Deployment

Virtual Servers in AWS & Apache/Nginx Web Server Setup

Module 1: AWS EC2 Fundamentals

Amazon Elastic Compute Cloud (Amazon EC2) provides secure, resizable compute capacity in the cloud. It acts as a virtual server (instance) in the AWS cloud environment, eliminating the need to invest in physical hardware. You can launch instances in minutes, scale them dynamically, and pay for only what you consume.

- **AMI (Amazon Machine Image):** A pre-configured template containing the operating system, application server, and system packages required to boot the server instance.
- **Elasticity:** The architectural capability of increasing or decreasing computing capacity dynamically in response to active traffic requirements.
- **Instance Types:** Pre-configured combinations of CPU, RAM, and network capabilities tailored to specific software applications.

Module 2: Deploying a Static Website on EC2

Deploying a static HTML page on an EC2 instance is a foundational DevOps exercise. It introduces server configuration, package installation, service management, and public network accessibility.

Step-by-Step Implementation Guide

- **1. Launch Instance:** Select the Amazon Linux 2023 or Ubuntu AMI. Choose the free-tier eligible instance type (e.g. t2.micro or t3.micro).
- **2. Firewall Configuration:** In the Security Group rules, add an Inbound Rule for 'HTTP' on Port 80 from source 'Anywhere (0.0.0.0/0)' to allow public web requests. Also allow 'SSH' (Port 22) for administration.
- **3. Connect to Instance:** Open Git Bash and run the SSH command with your private key pair: `ssh -i path/key.pem ec2-user@public-ip-address`
- **4. Install Web Server:** Install the Apache HTTP web server package using the local package manager: `sudo yum update -y && sudo yum install httpd -y`
- **5. Manage the Service:** Start the HTTP service and enable it to launch automatically on system reboot: `sudo systemctl start httpd && sudo systemctl enable httpd`
- **6. Upload Website Index:** Write your custom HTML page to the default document directory: `cd /var/www/html && echo '<h1>Web Server Running</h1>' | sudo tee index.html`
- **7. Test Page Access:** Copy the public IPv4 address of the EC2 instance and enter it into a browser. The static web page should render immediately.